Memory Is the Defence: Physics-Gated Continuum-Memory Receivers for Poisoning-Resilient Test-Time Learning

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***Abstract*—Neural receivers that learn at test time adapt to the channel from the pilots of every coherence block, so the pilot stream is a training set that an adversary can write to: a perturbation shaped through the receiver's own adaptation step destroys online and meta-learned DeepSIC receivers, and meta-learning retains the poison. This paper proposes Citadel-Rx, a DeepSIC receiver whose adaptable parameters are kept at three nested time-scales: a fast level fitted on every block and two slow levels updated every 8 and 32 blocks through a gated average of the faster states. Three label-free physics-consistency gates, each a ratio against a model-based reference on the same block (linear-model residual, CRC pass rate, channel innovation), decide what may be consolidated; a rejected block is reverted and its data are detected with the slow memory. For the algorithm as implemented, the displacement that a bounded attack causes in the slowest level over W blocks is proved to be bounded by the fast level's displacement times $1-(1-\eta^{(K)})^{\lceil W/C_K\rceil}$, and the bound is verified with measured constants. In 1 926 recorded runs over four channel families, 4×4 and 8×8 systems, three Doppler rates and five attackers, Citadel-Rx keeps its clean symbol error rate at every budget from -10 to $+10$ dB signal-to-interference ratio (attack-window SER 0.026–0.034 against 0.28–0.97 for online DeepSIC and 0.24–0.86 for Meta-DeepSIC, all paired differences significant after Holm correction), rejects 86–99% of poisoned blocks with at most 1.7% false rejections, and without attack halves the symbol error rate of online DeepSIC at 10 dB. The limits of the defence are established as well: a floor near 10^{-3} where the clean error rate is 10^{-4}, and no clean gain over LMMSE on a nearly diagonal channel.**

***Index Terms*—Deep receivers, test-time learning, adversarial pilot poisoning, meta-learning, continual learning, physics-informed gating, MIMO detection, 6G security.**

I. Introduction

DEEP receivers replace or augment the model-based blocks of a MIMO detector with neural networks trained on data [1]–[4]. Since a network trained in simulation meets a different channel in the field, the strongest of them adapt at test time: online DeepSIC [5] re-trains its per-user networks on the pilots of every coherence block, and Meta-DeepSIC [6] meta-learns an initialisation from the pilots of past blocks so that the per-block fit converges faster. Their adaptation data are the only labelled symbols the receiver sees: a few hundred pilot bits per user per block.

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Code, configuration files, all 1 926 run records, the statistics and the scripts that regenerate every table, figure and number of this paper are available at https://github.com/abraheemrashid99-sys/Citradel-Rx; the COST2100 traces are those released with [6].

TABLE I
Positioning by method class. The attack surface is what the adversary can write to; –: not addressed.

Class	Adapts	Memory	Attack surface / defence
LMMSE	no	–	pilot estimate / –
Online DeepSIC [5]	every block	last block	pilots / –
Meta-DeepSIC [6]	every block	pilot buffer	pilots, persistent / –
Drift-detection receiver	on drift	last block	pilots / adapt less
Robust TTA, vision [8], [9]	every batch	batch statistics	batch / robust statistics
Byzantine aggregation [10]	–	–	gradients / needs many vectors
Nested, continuum memory [13], [14]	every token	multi-rate	input / –
Citadel-Rx (this paper)	every block	three rates, gated	pilots / physics gates, certified retention

The security problem addressed here is the following. An adversary within range of the base station adds a perturbation to the received pilot part of a block, with an energy budget expressed as a signal-to-interference ratio (SIR) over the pilots. Wu *et al.* [7] showed that a perturbation shaped through an unrolled copy of the receiver's adaptation step raises the post-adaptation symbol error rate (SER) of online DeepSIC by an order of magnitude, that a perturbation crafted on a surrogate receiver still transfers, and that Meta-DeepSIC, which keeps the poisoned pilots in its support set, does not recover after the attack stops. They call for receivers that are secure by construction; none has been reported. Four properties of the problem shape the design and the evaluation. The receiver cannot verify its adaptation data, because the pilots are the labels; what it can verify is physical consistency: whether its decisions fit a linear channel model, whether the CRC of the decoded data passes, and whether the estimated channel moved as a channel does. The harm of poisoning lies not in the attacked block alone but in what the receiver keeps, so a defence must protect the memory and not only the decision. An attacker who knows the defence will try to pass its tests, so the tests must be tied to the attacker's own objective. And the value of a defence can only be established against the attacks of the literature, under the same channel realisations as the baselines, over many seeds, with the clean cost of the defence reported alongside its benefit.

Prior work addresses these properties separately (Table I). Test-time adaptation in vision has its own poisoning problem [8], and its defences replace batch statistics or gradients by robust estimates [9]; they assume i.i.d. batches and, as shown below, do not transfer to a receiver that sees one correlated

block at a time. Byzantine-robust aggregation [10] needs many gradient vectors per step, which a receiver with one pilot block does not have. Certified defences against data poisoning [11] bound the loss of a convex learner under a fraction of corrupted points; here the setting is sequential, the learner is a network, and the corruption is a bounded perturbation of every point in a block. Corruption-robust online learning [12] supplies the additive accounting of Section IV. Nested learning [13] and Titans [14] keep memory at several update frequencies but do not decide what enters the slow memory, and a single test-time memory is, as the ablation shows, as vulnerable as online DeepSIC. No published receiver keeps memory at several rates, gates what enters the slow rates on physics the attacker cannot forge without giving up its objective, and carries a retention guarantee checked against measurements.

The contributions of this paper are as follows.

1) **A three-timescale receiver** (Section III): a fast level with a delta-rule learner and a surprise-generated step size, slow levels at periods aligned with the coherence structure of the channel, gated consolidation, a proximal anchor that pulls the fast level toward the slow projection, and a gated fallback that detects a rejected block with the uncorrupted memory.
2) **Relative physics gates** (Section III-D): three label-free tests whose statistics are ratios against a model-based reference computed on the same block, so that a channel that merely degrades does not look like a poisoned adaptation. The CRC gate's threshold is derived, not tuned, from a target excess SER (Proposition 1), and a relative slack keeps it meaningful at 10^{-3} operating points.
3) **A retention certificate for the implemented algorithm** (Section IV): the exposure of the slowest level to a bounded attack is divided by its period and multiplied by the miss rate of the gate (Theorem 1), with a regret decomposition in which the adversary's term is additive (Theorem 2). Every constant is measured and the bound is drawn against paired adversarial/clean memory trajectories.
4) **A reproducible, paired evaluation** (Sections V–VI): seven receivers, five attackers, four channel families, two system sizes, three Doppler rates and three to five seeds; bootstrap intervals and Holm-corrected Wilcoxon tests on per-block differences paired on the channel realisation; every number regenerated from 1 926 recorded runs by one script each.

The evaluation also establishes where the defence does *not* help: at clean SER near 10^{-4} the attacked window sits near 10^{-3}, because the few poisoned blocks that pass the gate dominate; on a nearly diagonal channel the clean gain over LMMSE vanishes; and the first calibration of the relative CRC slack raised false rejections to 15–20% before it was corrected.

Section II presents the system and threat model, Section III the receiver, Section IV the guarantees, Section V the experimental setup, Sections VI and VII the results and their scope, and Section VIII concludes.

TABLE II
SIMULATION SETUP, READ FROM THE CONFIGURATION FILES AND THE CODE.

Item	Value
System	$U \times N = 4 \times 4$ (8×8 in Section VI-I), QPSK, SNR 10 dB (swept 4–16 dB)
Block	$N_p = 100$ pilot symbols (200 bits) and $N_d = 25\,000$ data symbols per user
CRC	CRC-8 with the 3GPP polynomial [22]; $L = 32$ symbols per sub-block; $n = 781$ sub-blocks per block
Channel	SED $[\boldsymbol{H}]_{ij} = e^{-\lvert i-j\rvert}$ with per-user cosine fading (periods 51, 39, 33, 21 blocks; depth 0.8) [5]; Jakes block correlation $\rho = J_0(2\pi f_D T_b)$ with $f_D T_b \in \{0.01, 0.05, 0.1\}$ [20]; $\tanh$ PA nonlinearity; COST2100 traces released with [6] [21]
Run	100 blocks (400 for A3); attack on blocks 30–69; seeds 0–2 (0–4 on the headline cells)
DeepSIC	$Q = 3$ iterations; per-user MLP with $64\cdot$order hidden units and ReLU [5]
Adaptation	Adam, learning rate $5 \cdot 10^{-3}$, 300 epochs per block on the pilots; joint pre-training at 2–16 dB
Attacker	PGD with 20 steps of size $0.25\,\epsilon$ through 2 unrolled adaptation steps [7], [16]

II. System Model and Threat Model

This section fixes the uplink, the receiver under attack, the attacker and the metrics by which every method is judged.

A. Uplink and Adaptive Receiver

U single-antenna users transmit QPSK symbols to a base station with N antennas. Time is divided into coherence blocks $b = 1, 2, \ldots$; within block b the channel $\boldsymbol{H}_b \in \mathbb{C}^{N\times U}$ is constant and the received vector at symbol slot n is

$$\boldsymbol{y}_n = f(\boldsymbol{H}_b \boldsymbol{s}_n) + \boldsymbol{w}_n, \qquad \boldsymbol{w}_n \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}), \tag{1}$$

with f the identity (linear channel) or the memoryless power-amplifier nonlinearity $f(x) = \tanh(0.5x)$ applied per antenna, as in [6]. Each block carries N_p pilot symbols per user, known to the receiver, followed by N_d data symbols framed in CRC-protected sub-blocks of L symbols (Table II). Four channel families are used (Section V).

The receiver is DeepSIC [5]: Q soft-interference-cancellation iterations, each a per-user multilayer perceptron that maps the received vector and the other users' current soft decisions to a posterior over the user's symbol. Its parameters θ are adapted on the pilots of every block,

$$\theta_b = \Phi(\theta_{b-1}; \boldsymbol{Y}_b), \tag{2}$$

by T steps of a first-order optimiser on the pilot cross-entropy, and then detect the data of the block. Online DeepSIC starts every block from θ_{b-1}; Meta-DeepSIC starts from an initialisation meta-learned by MAML [15] on a buffer of past pilot blocks, refreshed every five blocks [6].

B. Threat Model

The adversary adds $\boldsymbol{E}_b$ to the received pilot matrix $\boldsymbol{Y}_b \in \mathbb{C}^{N\times N_p}$ of block b, with $\lVert\boldsymbol{E}_b\rVert_F \le \epsilon_b$ set by the SIR over the pilot block,

$$\epsilon_b = \lVert\boldsymbol{Y}_b\rVert_F \; 10^{-\mathrm{SIR}/20}, \tag{3}$$

so that SIR 0 dB spends as much energy as all U users together; the data part is not touched. The attacker knows the pilots, which are standardised, and observes the channel

through its own receiver. Following and extending [7], four attackers are considered.

- *A1, availability.* Projected gradient ascent [16] on the post-adaptation data loss, differentiated through an unrolled copy of the victim's adaptation map (2) (white-box), or crafted on a surrogate online DeepSIC and injected (transfer). A receiver without an adaptation map (LMMSE) is attacked through the surrogate.
- *A2, targeted drift.* A1 with an objective that raises one user's loss while keeping the others' low, on a Jakes channel, so that the poison resembles a plausible channel change.
- *A3, boiling frog.* Many blocks of small perturbations (5% of the pilot energy) that push the receiver toward a spoofed channel basin while keeping the SER inside any tolerance a detector might use.
- *Gate-aware.* A1 with a penalty on the defence's gate statistics in the attacker's objective: an attacker who knows the defence and tries to pass it.

A *data-jamming control* spends the same energy on the data part of the block instead of the pilots; it separates persistent poisoning from plain interference.

C. Metrics

The SER is the fraction of symbol slots in which at least one user is wrong, the metric of [5]–[7], reported as a mean over the attack window (blocks 30–69), before it (pre) and after it (post). The recovery time is the number of blocks after the attack until the 3-block moving average of the SER returns within 10% of the pre-attack mean (30: not within the run). The gate is characterised by its rejection rate of poisoned blocks and its false-rejection rate of clean blocks. The degradation Δ_{dB} is the extra SNR needed under attack to reach the clean SER [7]. The paired displacement $D_k(W) = \left\|\theta^{(k)}_{\mathrm{adv}} - \theta^{(k)}_{\mathrm{clean}}\right\|$ of memory level k is measured between an attacked and a clean run on the same channel. Every number of the paper is regenerated from the recorded runs by one script each; finished records are skipped on re-run, so every stage of the study is resumable.

III. Citadel-Rx

The receiver splits adaptation by time-scale: a fast level fits every block; slow levels are updated at the periods of the coherence structure of the channel, and only with what the gates accept. Fig. 1 shows the architecture.

A. Levels, Periods and the Fast Learner

The adaptable parameters $\theta \in \mathbb{R}^d$ (all DeepSIC weights, $d = 15\,628$ for 4×4) are kept at $K = 3$ levels $\theta^{(1)}, \theta^{(2)}, \theta^{(3)}$ with periods $C = (1, 8, 32)$ blocks, each a full copy. The fast level is adapted on every block by $T = 300$ steps of

$$\theta \leftarrow \theta - \eta_b \delta\, \theta\, \boldsymbol{\phi}\boldsymbol{\phi}^{\top} \quad \text{(delta-rule decay, per weight matrix)}, \tag{4}$$

$$\theta \leftarrow \theta - \eta_b \lambda(\theta - a_b) \quad \text{(proximal pull to the anchor)}, \tag{5}$$

$$\theta \leftarrow \mathrm{Adam}\big(\theta, \nabla_\theta \ell_b(\theta)\big), \tag{6}$$

where ℓ_b is the pilot cross-entropy of the DeepSIC stage being adapted (stages are adapted sequentially as in [5]), $\boldsymbol{\phi}$ is the normalised mean input of the stage (the direction along which the delta rule forgets), $\delta = 0.1$, $\lambda = 0.5$, and $a_b = \sum_{k \geq 2} w_k \theta^{(k)}_{b-1}$ with $w = (\frac{1}{2}, \frac{1}{2})$ is the *anchor*, the slow projection. The step size is generated by the receiver's own surprise: with $S_b = \|\nabla_\theta \ell_b(\theta_{b-1})\|^2$ and $\bar{S}_b$ its exponential moving average over past blocks, seeded by the median of the first five,

$$\eta_b = \eta_{\max} \frac{2 r_b}{1 + r_b}, \qquad r_b = \mathrm{clip}\big(S_b / \bar{S}_b, \tfrac{1}{8}, 8\big), \tag{7}$$

so that an unsurprising block receives a small step and a surprising one at most $2\eta_{\max}$. The rate is bounded, which the analysis requires.

B. Gated Consolidation

Every C_{k+1} blocks, level $k{+}1$ receives the gated mean of the level-k states buffered since its last update, each weighted by the gate verdict $\gamma_j \in [0, 1]$ of its block (level 1) or by the mean verdict of its period (level 2):

$$\theta^{(k+1)} \leftarrow \theta^{(k+1)} + s\Big(\frac{\sum_j \gamma_j \theta^{(k)}_j}{\max(1, \sum_j \gamma_j)} - \theta^{(k+1)}\Big), \qquad s = \eta^{(k+1)} \min\big(1, \textstyle\sum_j \gamma_j\big), \tag{8}$$

with $\eta^{(2)} = 1$ (level 2 is a running mean of the fast level) and $\eta^{(3)} = 0.5$. Eq. (8) is a convex combination (Lemma 1): a level never leaves the convex hull of what it has been shown. The fast level is anchored continuously by (5) rather than reset at consolidation instants as in [13]; Section VI-H shows that this choice is the source of the clean gain.

C. Rejection, Reversion and Fallback

If $\gamma_b < \frac{1}{2}$ the block is rejected: the fast update is reverted, $\theta^{(1)}_b = \theta^{(1)}_{b-1}$; the reverted state enters the buffer with weight γ_b; and the data of the block are detected with the anchor, the memory the attacker has not reached. Reversion protects the memory; the fallback protects the attacked block itself (Section VI-H: without it the attack-window SER is 0.245, with it 0.028). The remaining settings, all read from the configuration file, are $\eta_{\max} = 5 \cdot 10^{-3}$, a surprise EMA weight of 0.9 with a five-block warm-up, gate thresholds $\tau_1 = 1.25$ and $\tau_3 = 4$ (the latter floored by the $\chi^2_{0.999}$ quantile per degree of freedom), a soft step of width 0.1τ around every threshold, and thresholds tightened by up to 30% under surprise.

D. Relative Physics Gates

Every statistic is a ratio against a reference computed on the same block from the received data and no labels: the LMMSE detector built on the least-squares pilot channel estimate $\hat{\boldsymbol{H}}_b$, and the detector obtained with the anchor parameters; the reference is the better of the two. Ratios are normalised by their exponential moving average over accepted blocks and compared with a soft threshold, tightened by up to 30% when the surprise ratio r_b exceeds one. Fig. 2 gives the decision inside one block.

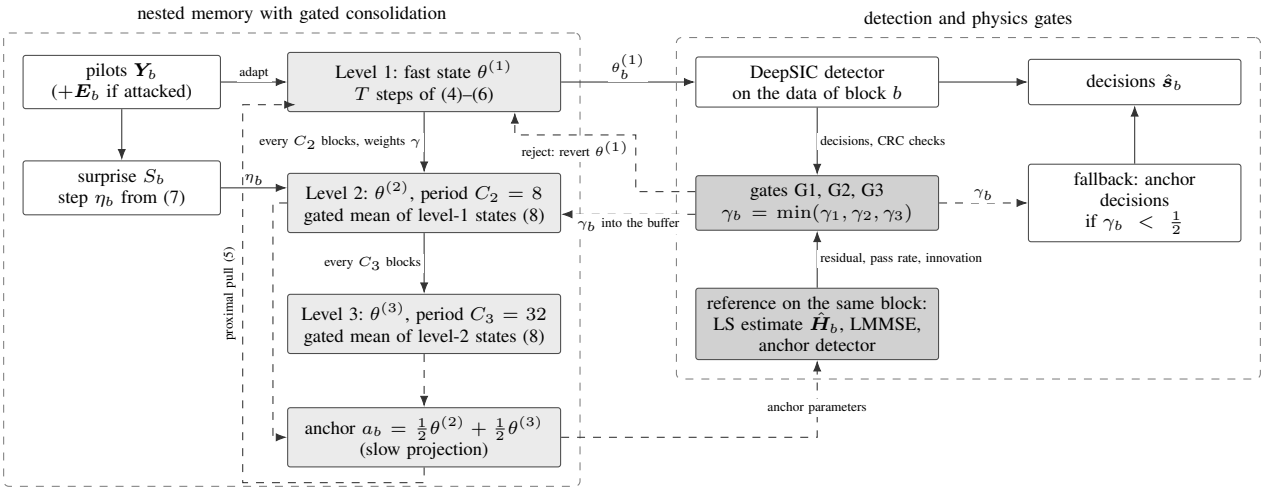


Fig. 1. Citadel-Rx. Solid arrows carry parameters and decisions; dashed arrows carry the gate verdict and the slow projection fed back. Level $k{+}1$ receives the gated mean of level-k states every C_{k+1} blocks (8); the anchor pulls the fast level on every step (5); a rejected block is reverted and its data are detected with the anchor.

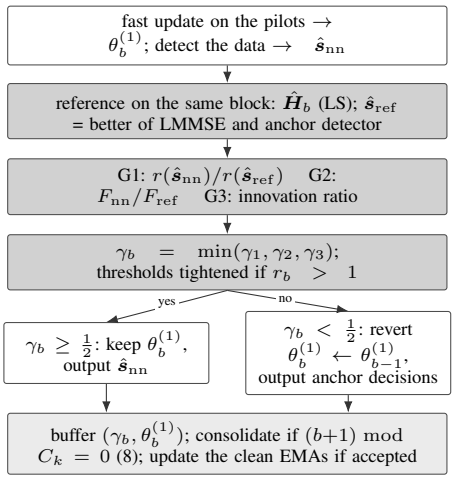


Fig. 2. Decision inside one block. Light grey: memory operations; dark grey: gate computations.

G1, model residual. With $r(\hat{\boldsymbol{s}}) = \frac{1}{N_d}\sum_n \left\| \boldsymbol{y}_n - \hat{\boldsymbol{H}}_b \hat{\boldsymbol{s}}_n \right\|^2$, the statistic is $r(\hat{\boldsymbol{s}}_{\mathrm{nn}})/r(\hat{\boldsymbol{s}}_{\mathrm{ref}})$. A poisoned adaptation produces decisions inconsistent with the linear model. Both terms contain the same noise and model-mismatch energy, so a PA nonlinearity compresses the ratio toward one and cannot raise a false alarm (Section IV).

G2, CRC pass rate. The fraction F_{nn} of CRC sub-blocks that pass with the fast level, over the fraction F_{ref} with the reference, against the threshold τ_2 of Proposition 1. This gate is tied to the availability attacker's own objective: an attack that raises the SER lowers the pass rate.

G3, channel prior. The innovation energy of the LS channel estimate against its AR(1) prediction, relative to its running level on accepted blocks and floored by the χ^2 quantile of the degrees of freedom. It targets physically implausible jumps and is the only gate not fooled by an attacker who keeps the SER low.

IV. Guarantees

The guarantees are stated for the algorithm as implemented; the full proofs are given in the theory document of the repository. Let $\Phi(\theta; \boldsymbol{Y}, a)$ denote the block adaptation map (the fast state after one block, given the pilots and the anchor), and consider paired runs on the same clean sequence with and without an attack, with displacements $d_b = \left\| \theta_b^{(1),\mathrm{adv}} - \theta_b^{(1),\mathrm{clean}} \right\|$ and $D_k(b)$ for level k.

Lemma 1 (consolidation is a convex combination)**.** *Eq.* (8) *writes the new level-*$(k{+}1)$ *state as a convex combination of its old state and the buffered level-*k *states, with total weight* $s \leq \eta^{(k+1)}$ *on the latter. Consequently, if the paired runs have equal verdicts,* $D_{k+1} \leq \max\{D_{k+1}^{\mathrm{old}}, \max_j D_k(j)\}$*, and by induction* $D_k(b) \leq \max_{j \leq b} d_j$ *for every* $k \geq 2$*.*

Theorem 1 (retention certificate)**.** *Let an attack occupy a window of* W *blocks with equal gate verdicts in the paired runs, let* $n(W) \leq \lceil W/C_K \rceil$ *be the number of level-*K *consolidations inside it, and let* $\Delta_f(W) = \max_{\mathrm{window}} d_b$ *be the largest paired fast-level displacement. Then*

$$D_K(W) \leq \Delta_f(W)\big(1-(1-\eta^{(K)})^{n(W)}\big) \leq \eta^{(K)}\Delta_f(W)n(W). \tag{9}$$

If moreover $\|\Phi(\theta; \boldsymbol{Y}+\boldsymbol{E}, a) - \Phi(\theta; \boldsymbol{Y}, a)\| \leq G_\Phi \|\boldsymbol{E}\|$ *and* Φ *is* L_θ*- and* L_a*-Lipschitz in* θ *and* a *with* $L = L_\theta + L_a < 1$*, then* $\Delta_f(W) \leq G_\Phi \epsilon \min\{1, (1-L_\theta)n_W\}/(1-L)$*, where* $n_W = \rho_{\mathrm{blk}}(1-p^{\mathrm{rej}})W$ *counts the attacked blocks the gate accepts; if only* $L \leq 1$ *holds (non-expansive map),* $\Delta_f(W) \leq G_\Phi \epsilon\, n_W$*.*

Proof sketch. Level K consolidates $n(W)$ times in the window; each time, by Lemma 1, $D_K \leftarrow (1-s)D_K + s\Delta_f$ with $s \leq \eta^{(K)}$, which iterated from $D_K = 0$ gives (9). For the second part, the triangle inequality on Φ gives $d_b \leq L_\theta d_{b-1} + L_a \max_{k \geq 2} D_k(b-1) + G_\Phi \epsilon \chi_b$ with $\chi_b = 1$ on attacked accepted blocks; the lemma and the geometric sum give the bound, and with $L \leq 1$ the recursion is non-expansive and adds at most $G_\Phi \epsilon$ per accepted attacked block. □

TABLE III
CONSTANTS OF THEOREM 1 MEASURED BY FINITE DIFFERENCES ON CLEAN BLOCKS OF THE A3 CONFIGURATION: MAXIMUM OVER 10 BLOCKS AND 3 RANDOM PROBES, MEAN IN PARENTHESES. SCRIPT: CERTIFICATE_CHECK.PY.

λ	G_Φ	L_θ	L_a	$L_\theta + L_a$	ϵ (A3)
0.5 (default)	5.21 (3.13)	0.86 (0.76)	0.35 (0.32)	1.20	1.05
1	1.54 (1.43)	0.54 (0.52)	0.54 (0.53)	1.08	1.11
2	2.22 (1.97)	0.40 (0.35)	0.82 (0.80)	1.22	1.11
4	2.21 (2.06)	0.38 (0.29)	1.01 (0.99)	1.39	1.11

The first part needs no regularity of the fast learner: the exposure of the slow level is divided by its period whatever the fast level does. The second part identifies where the gate enters, through n_W, and why rejected blocks inject nothing: they are reverted. The form $\eta^{(K)}\rho\,\epsilon\,G(1-p^{\mathrm{rej}})W/C_K$ holds with $G = G_\Phi/(1-L)$.

Theorem 2 (corruption-robust dynamic regret)**.** *Let ℓ_b be G_ℓ-Lipschitz in θ and $R_T(u) = \sum_b \ell_b(\theta_b) - \ell_b(u_b)$ the dynamic regret against a comparator sequence. Under the hypotheses of Theorem 1, $R_T^{\mathrm{adv}}(u) \le R_T^{\mathrm{clean}}(u) + G_\ell\, T\, \Delta_f(T)$; without the anchor ($L_a = 0$), $R_T^{\mathrm{adv}}(u) \le R_T^{\mathrm{clean}}(u) + G_\ell G_\Phi \epsilon\, n_T/(1-L_\theta)$; and a detector that uses level k pays at most $G_\ell \Delta_f(T)\min\{T, \eta^{(k)}T^2/(2C_k)\}$.*

The adversary's contribution is additive to the clean regret, linear in the number of accepted corrupted blocks and divided by the level period. No rate is claimed for the clean term of the full-MLP receiver, which is reported empirically; for a last-layer-adaptable (convex) variant the $O(\sqrt{T(1+P_T)})$ bound of [17], [18] applies with the bounded step (7).

Proposition 1 (CRC gate)**.** *With i.i.d. symbol errors of probability p in a sub-block of L symbols and an r-bit CRC whose syndrome is uniform on error, $P(\mathrm{pass} \mid p) = (1-p)^L + 2^{-r}(1-(1-p)^L)$. Inverting at F_{ref} gives p_{ref}, and the rule "reject if $F_{\mathrm{nn}}/F_{\mathrm{ref}} < \tau_2$" with*

$$\tau_2 = \Big(\frac{1-p_{\mathrm{ref}}-\delta}{1-p_{\mathrm{ref}}}\Big)^L, \qquad \delta = \max\{\delta_{\min}, \rho\, p_{\mathrm{ref}}\}, \quad (10)$$

rejects exactly when the fast level's error probability exceeds the reference's by more than δ, up to the Hoeffding fluctuation $2e^{-2nt^2}$ of the pass fraction over n sub-blocks [19].

With $n = 781$ the gate is deterministic for practical purposes against any attacker that raises the SER by more than δ, and an attacker that does not has caused less than δ of harm. The absolute rule ($\delta \equiv 0.01$) is $30\times$ the clean SER at a $3\cdot 10^{-4}$ operating point; the relative rule with $\rho = 1$ and $\delta_{\min} = 2\cdot 10^{-3}$ (five standard deviations of the pass fraction) follows it.

The residual gate is insensitive to model mismatch for the same reason: writing $\boldsymbol{y}_n = \hat{\boldsymbol{H}}\boldsymbol{s}_n + \boldsymbol{\nu}_n + \boldsymbol{m}_n$ with $\boldsymbol{m}_n$ the mismatch term, the residual of a decision with error pattern $\boldsymbol{e}_n$ is $\mathrm{mean}\left\|\hat{\boldsymbol{H}}\boldsymbol{e}_n\right\|^2 + \mathrm{mean}\,\|\boldsymbol{\nu}_n + \boldsymbol{m}_n\|^2 + O(N_d^{-1/2})$, so the ratio of the two residuals is monotone in the excess decision-error energy of the fast level and the mismatch only compresses it toward one.

Table III gives the measured constants. $L_\theta + L_a$ stays near one at every proximal weight: what the fast map forgets about its own past ($1-L_\theta$) it inherits from the anchor (L_a), because the proximal step drives θ toward the anchor. The joint system is non-expansive rather than contractive, so the part of Theorem 1 claimed for the default receiver is the structural bound (9) with the measured Δ_f, checked in Section VI-G; the contraction branch is stated for completeness and is not certified by these probes. The same table shows that $\lambda = 1$ cuts the input sensitivity G_Φ of the fast level from 5.2 to 1.5: a stronger prior makes the per-block fit less steerable by the pilots.

V. EXPERIMENTAL SETUP

A fair test needs strong attackers, the receivers of the literature, several channel families and many seeds, with every receiver seeing the same realisation. The experiments are: clean SER against SNR (4–16 dB) on the four channel families (E2); A1 white-box and transfer at SIR -10 to 10 dB with the jamming control on the SED channel (E3), across SNR (E3snr), on the tanh PA channel and on COST2100 (E3nl, E3cost); A2 on Jakes fading with and without G3 (E4); A3 over 300 blocks with $C_K \in \{8, 32, 128\}$ and a paired clean run (E5); the gate-aware attacker with penalty weights 5, 50 and 500 against gate subsets (E7); the 8×8 system (E8); Jakes fading at $f_D T_b = 0.01$, 0.05 and 0.10 (E9); and sixteen ablation variants. All use 3 seeds; E3 and the 10 dB clean cells use 5.

Baselines. LMMSE on the least-squares pilot estimate (no adaptation; the model-based reference); online DeepSIC [5]; Meta-DeepSIC [6]; a drift-detection receiver that adapts only when a z-test on the pilot loss fires (the "adapt less" strategy); a MedBN-style receiver that replaces the adaptation gradient by a median of means over four pilot groups [9]; and a single-memory Titans receiver, the same delta-rule fast learner and surprise rate with $K = 1$ and no gates [14], which isolates the memory hierarchy from the fast rule. All DeepSIC-type receivers share the architecture, the pre-training and the 300 adaptation steps per block of Table II.

Protocol. Every receiver is run on the same channel realisation, noise and attack draw for a given seed (paired design); the attacker re-crafts its perturbation for each victim. Means over seeds are reported, with 95% bootstrap confidence intervals over the (seed, block) pairs (10 000 resamples [23]) and two-sided Wilcoxon signed-rank tests [24] on the paired per-block differences, Holm-corrected within each table [25], for the main cells. The attack window, recovery criterion and budget were fixed before the sweeps; the one change made after seeing results, the relative CRC slack, is reported with its first, failed calibration (Section VI-J).

VI. RESULTS

The receivers are compared under the availability attack, then from the gate's side, across SNR, with significance tests, under the stress attackers, against the certificate, in ablation, and on other channels, a larger system and faster fading.

A. Main Comparison

Table IV and Figs. 3–4 contain the main result. Under the white-box attack, every baseline is destroyed at SIR $\le$ 0 dB

TABLE IV

SER IN THE ATTACK WINDOW (BLOCKS 30–69) UNDER A1 AT SIR $-10/-5/0/5/10$ DB ON THE 4×4 SED CHANNEL AT 10 DB; MEAN OVER SEEDS (5 FOR ONLINE DEEPSIC, META-DEEPSIC AND CITADEL-RX, 3 OTHERWISE). PRE: BLOCKS 0–29. POST AND RECOVERY (REC., BLOCKS; 30: NOT WITHIN THE RUN) AT SIR 0 DB. LMMSE HAS NO ADAPTATION MAP; ITS ATTACKED ROW IS THE TRANSFER ONE. SCRIPTS: `SWEEP.PY E3`, `ANALYZE.PY`.

Receiver	Pre	White-box A1	Post	Rec.	Transfer A1	Rec.	Jamming 0 dB
LMMSE	0.056	–	–	–	0.873 / 0.740 / 0.447 / 0.149 / 0.064	0	0.892
Online DeepSIC	0.071	0.973 / 0.938 / 0.772 / 0.475 / 0.277	0.088	11	0.352 / 0.357 / 0.347 / 0.295 / 0.210	10	0.855
Meta-DeepSIC	0.062	0.861 / 0.805 / 0.593 / 0.400 / 0.241	0.186	30	0.195 / 0.201 / 0.213 / 0.211 / 0.175	14	0.851
Drift-detection	0.077	0.975 / 0.942 / 0.776 / 0.459 / 0.265	0.097	10	0.345 / 0.359 / 0.337 / 0.277 / 0.199	7	0.858
MedBN-style	0.065	0.933 / 0.853 / 0.738 / 0.549 / 0.397	0.204	30	0.724 / 0.641 / 0.544 / 0.423 / 0.296	30	0.858
Single-memory Titans	0.068	0.975 / 0.939 / 0.788 / 0.486 / 0.276	0.075	6	0.363 / 0.368 / 0.354 / 0.290 / 0.200	6	0.861
Citadel-Rx (proposed)	**0.040**	**0.026 / 0.025 / 0.028 / 0.030 / 0.034**	**0.029**	**0**	**0.025 / 0.025 / 0.027 / 0.030 / 0.037**	**0**	0.848

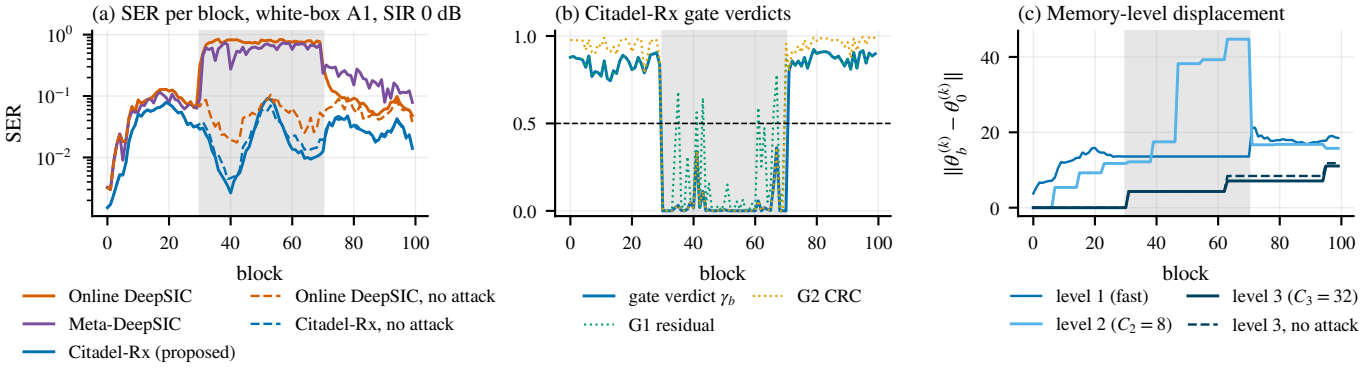


Fig. 3. One run (seed 0, white-box A1 at SIR 0 dB; shaded: attack window). (a) SER per block; dashed: the same receiver without attack. (b) Citadel-Rx gate verdicts: the CRC gate drops to zero on every poisoned block and the combined verdict rejects it. (c) Displacement of the three memory levels from their pre-trained value; level 2 absorbs the weights of the reverted fast states and returns, level 3 follows its no-attack trajectory. Script: `fig_paper.py`.

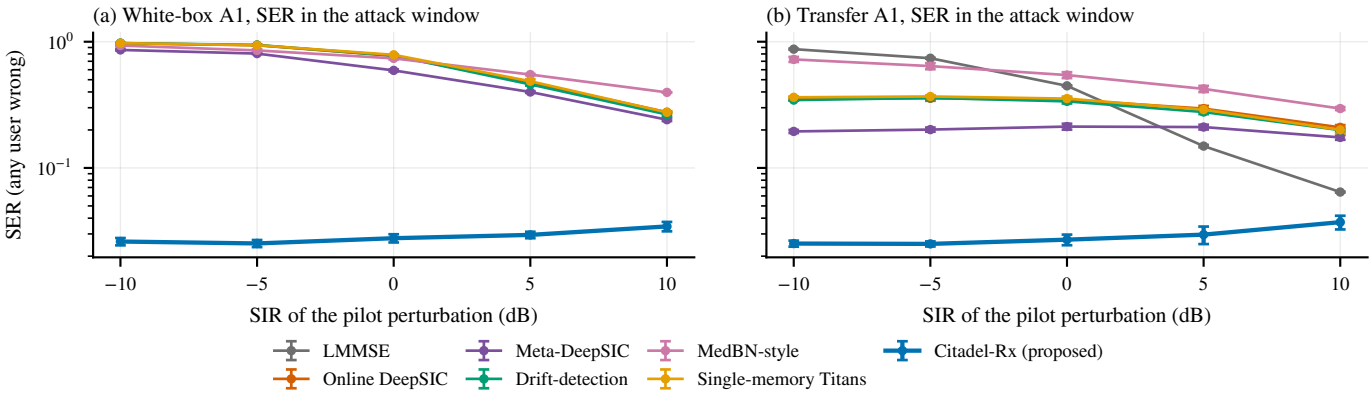


Fig. 4. SER in the attack window against the attacker's budget under (a) white-box and (b) transfer A1. Markers: seed means; bars: seed standard deviation. LMMSE appears in (b) only, since it has no adaptation map to attack through. Script: `fig_paper.py`.

(SER 0.59–0.98) and still loses 4–6× at 10 dB; Meta-DeepSIC and the MedBN-style receiver do not recover within 30 blocks (post-attack SER 0.19–0.20 against 0.06–0.07 before), reproducing the observation of [7]. The transfer attack is weaker against the baselines (0.20–0.36) but far from harmless, and it wrecks LMMSE at high budgets. The jamming control places the baselines' vulnerability in perspective: against every one of them, poisoning 200 pilot bits costs as much as jamming 50 000 data bits with the same energy (0.77 against 0.86 at 0 dB) and, unlike jamming, leaves persistent damage.

Against Citadel-Rx the attack is worthless at every budget and under both knowledge models. The attack-window SER (0.025–0.037) is at or below the receiver's own clean level in those blocks (Fig. 3a: the window covers the easy part of the fade, so the relevant comparison is with the dashed no-attack curve, from which the attacked run is indistinguishable), recovery is immediate, and the slow memory follows its no-attack trajectory (Fig. 3c). The attacker is reduced to jamming, which costs continuous power and leaves no trace. Before the attack, Citadel-Rx is 44% below online DeepSIC (0.040 against 0.071).

B. The Gate's Side

Fig. 5 views the same runs from the gate. The rejection rate of poisoned blocks is 0.98/0.98/0.97/0.95/0.86 under the white-box attack and 0.99/0.99/0.97/0.94/0.78 under the transfer attack at SIR $-10/-5/0/5/10$ dB, with false rejections of clean blocks between 0.000 and 0.017. Rejection

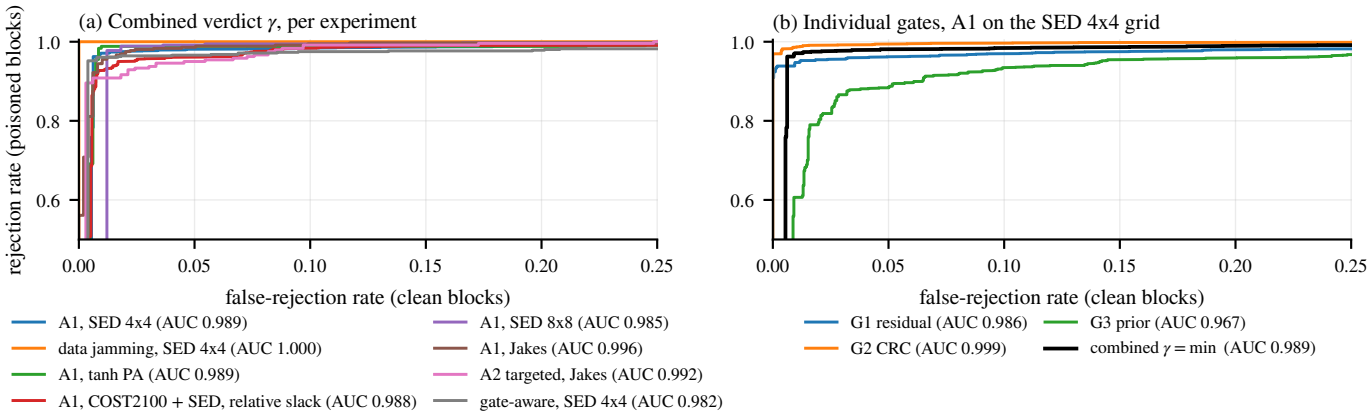


Fig. 5. Receiver operating characteristic of the gate verdict γ as a detector of poisoned blocks, pooled over seeds and budgets (axes zoomed to the upper-left corner): (a) combined verdict per experiment; (b) individual gates on the A1 grid. At the operating point $\gamma < \frac{1}{2}$, 94% of poisoned blocks are rejected at a 0.6% false-rejection rate on the A1 grid. Script: `gate_roc.py`.

falls with the attacker's budget because a weak perturbation does less damage and more often stays inside the CRC slack; the blocks it leaks cost little, which is why the SER in Table IV rises only from 0.026 to 0.034. Pooled over seeds and budgets, the combined verdict has AUC 0.989 under A1 (G1 alone 0.986, G2 alone 0.999, G3 alone 0.967) and 1.000 under data jamming, where the residual ratio is inverted (G1 alone 0.117) and the CRC gate carries the verdict: the gates are not interchangeable, and $\gamma = \min$ inherits the best of them.

C. Degradation Across SNR

Fig. 6 reports the degradation metric of [7]. For every baseline the attacked SER stays at 0.45–0.91 over the whole 4–16 dB grid, so the extra SNR needed to recover the clean SER exceeds the grid everywhere. For Citadel-Rx, $\Delta_{\mathrm{dB}} \leq 0$ from 4 to 10 dB and ≈ 0 at 12 dB with the relative slack (0.0062 attacked against 0.0059 clean); at 14 and 16 dB the attacked window sits at 0.0018 and 0.0009 against clean 0.0008 and 0.0001: the defence holds the receiver near 10^{-3} under attack but does not preserve a 10^{-4} operating point, because the 5–10% of poisoned blocks that pass the gate set the floor.

D. Significance and Seed Spread

Table V compares like with like: every pair of receivers sees the same channel, noise and attack draw, and the per-block differences are tested with Holm correction. Every difference in Tables IV and VI is significant at $p < 0.001$, and the intervals exclude zero by a wide margin. Over five training seeds, the attacked SER of Citadel-Rx is 0.026 ± 0.002, 0.028 ± 0.002 and 0.034 ± 0.003 at SIR -10, 0 and 10 dB (online DeepSIC 0.973 ± 0.003, 0.772 ± 0.013, 0.277 ± 0.005; Meta-DeepSIC 0.861 ± 0.009, 0.593 ± 0.011, 0.241 ± 0.006), and the clean SER at 10 dB is 0.0302 ± 0.0004 against 0.0590 ± 0.0008 and 0.0527 ± 0.0008: the seed spread is two orders of magnitude below the effect. The Jakes clean cell spreads for every receiver because the realised fade differs by seed (0.0059 ± 0.0028 against 0.0128 ± 0.0058 and 0.0115 ± 0.0050); the ranking is the same in all five seeds.

E. Clean Performance

Citadel-Rx costs less than nothing on three of the four channel families (Table VI): it is the best receiver at every SNR from 4 to 12 dB on the cosine, Jakes and nonlinear channels, halving the SER of online DeepSIC at 10 dB. Pre-training does not explain the gain (pre-trained online DeepSIC: 0.074 against 0.069 in the pre-attack blocks), nor does the fast rule alone (single-memory Titans: 0.053); Section VI-H attributes it to the proximal anchor, the slow memory acting as a prior for a fit from 100 pilot symbols. On COST2100, a nearly diagonal 4×4 channel on which LMMSE is close to optimal, Citadel-Rx matches LMMSE within 2–15% at 4–8 dB while every DeepSIC baseline is 1.5–2.3× worse than LMMSE. The clean gain is therefore channel-dependent, and nil where interference cancellation has nothing to cancel.

F. Targeted, Boiling-Frog and Gate-Aware Attackers

Table VII stresses the gates. The targeted attacker on a Jakes channel takes every user of online DeepSIC and Meta-DeepSIC down (window SER 0.44 and 0.35), while Citadel-Rx stays at its clean 0.010 with 82% rejection, with or without the prior gate. Against the residual gate alone, the gate-aware attacker with a large penalty weight cuts the rejection rate from 0.93 to 0.30 at the price of a weaker poison (SER $0.034 \rightarrow 0.074$): evading the residual test costs poisoning strength, but G1 alone is insufficient. Against G1 and G2 together, or all three, the best-effort attacker gains nothing (rejection 0.95–1.00, SER 0.024–0.027). The boiling-frog attacker passes the gates by construction (rejection 0) and buys one SER point over 300 blocks; its effect on the memory is the subject of the certificate.

G. The Certificate With Measured Constants

Fig. 7 draws (9) against the paired adversarial/clean trajectories of the slowest level ($\eta^{(K)} = 0.5$, $p^{\mathrm{rej}} = 0$). The bound holds at every W for every period: the largest ratio of measured displacement to bound is 0.94, 0.53 and 0.26 for $C_K = 8$, 32 and 128, with end-of-window Δ_f of 33.4, 17.2 and 15.9. It is nearly tight at $C_K = 8$, where the slow

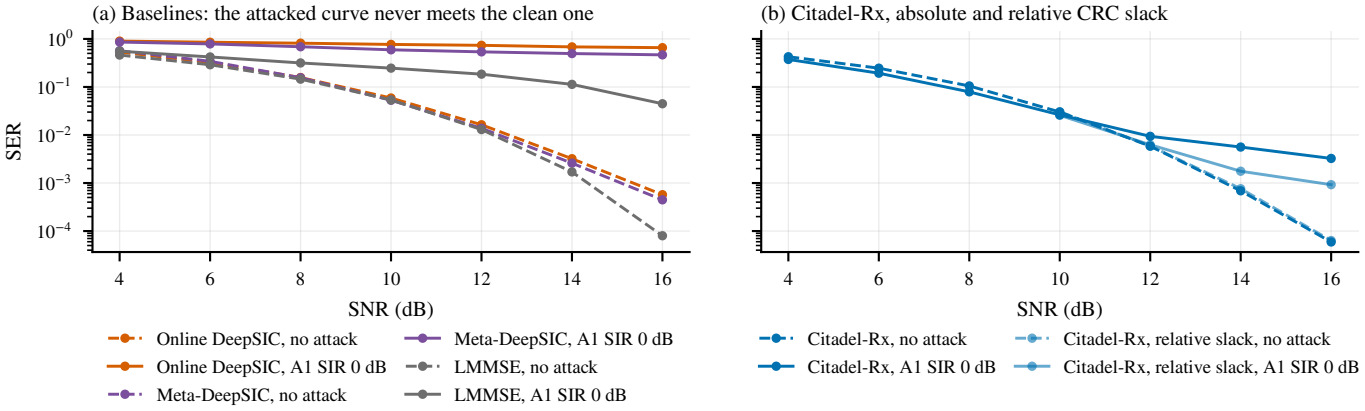


Fig. 6. SER against SNR without attack (dashed) and under white-box A1 at SIR 0 dB (solid), 3 seeds. (a) For every baseline the attacked curve never meets the clean one on the grid, so Δ_{dB} is unbounded. (b) Citadel-Rx with the absolute and the relative CRC slack; Δ_{dB} is the horizontal gap between the two curves. Script: `fig_paper.py`.

TABLE V

PAIRED COMPARISON ON (SEED, BLOCK) PAIRS: CITADEL-RX MINUS BASELINE, WITH 95% BOOTSTRAP INTERVAL AND HOLM-CORRECTED WILCOXON p (EVERY ENTRY $p < 0.001$). ATTACK CELLS USE THE WINDOW; CLEAN CELLS USE ALL 100 BLOCKS. FULL TABLES: `DOCS/STATS.MD`. SCRIPT: `STATS.PY`.

Citadel-Rx vs.	White-box A1, SIR 0 dB	Transfer A1, SIR 0 dB	Clean, cosine 10 dB	Clean, Jakes 10 dB
Online DeepSIC	-0.745 [-0.757, -0.731]	-0.320 [-0.331, -0.308]	-0.029 [-0.030, -0.028]	-0.0069 [-0.0075, -0.0063]
Meta-DeepSIC	-0.565 [-0.583, -0.546]	-0.186 [-0.198, -0.174]	-0.023 [-0.024, -0.021]	-0.0056 [-0.0061, -0.0051]
Drift-detection	-0.750 [-0.764, -0.734]	-0.310 [-0.325, -0.295]	-0.035 [-0.037, -0.033]	-0.0152 [-0.0166, -0.0139]
MedBN-style	-0.712 [-0.736, -0.686]	-0.517 [-0.539, -0.496]	-0.034 [-0.036, -0.032]	-0.0089 [-0.0097, -0.0080]
Single-memory Titans	-0.761 [-0.776, -0.745]	-0.328 [-0.342, -0.312]	-0.023 [-0.025, -0.021]	-0.0039 [-0.0046, -0.0032]
LMMSE	–	-0.420 [-0.432, -0.409]	-0.024 [-0.025, -0.022]	-0.0042 [-0.0049, -0.0035]

TABLE VI

SER WITHOUT ATTACK OVER 100 BLOCKS, 3 SEEDS (5 AT 10 DB ON THE COSINE AND JAKES CHANNELS). SCRIPTS: `SWEEP.PY E2`, `E2NL`, `E2COST`; `ANALYZE.PY`.

Receiver	Cosine 10 dB	Jakes 10 dB	tanh PA 10 dB	COST2100 8 dB
LMMSE	0.0537	0.0095	0.0562	**0.0041**
Online DeepSIC	0.0590	0.0128	0.0652	0.0096
Meta-DeepSIC	0.0527	0.0115	0.0595	0.0093
Drift-detection	0.0653	0.0206	0.0691	0.0140
MedBN-style	0.0644	0.0142	0.0605	0.0080
Single-memory Titans	0.0529	0.0092	0.0567	0.0102
Citadel-Rx (proposed)	**0.0302**	**0.0059**	**0.0331**	0.0047

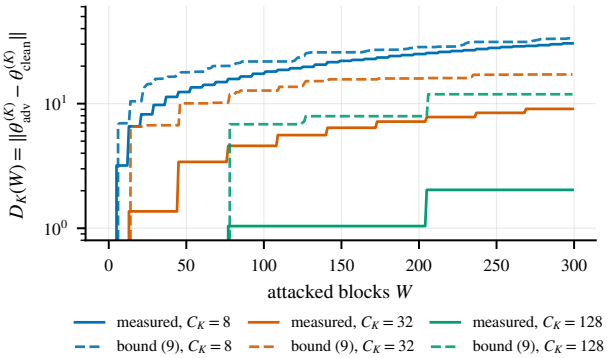


Fig. 7. Paired displacement of the slowest level under A3 (solid; 3 seeds) and the structural certificate (9) with the measured fast-level displacement (dashed) for $C_K = 8, 32, 128$ over 300 attacked blocks. Script: `certificate_check.py`.

level is a running average of the fast level, and the measured slopes of D_K (0.084, 0.029 and 0.0083 per block) scale as $1 : 0.35 : 0.10$ against the $1 : 0.25 : 0.0625$ of pure $1/C_K$; the

TABLE VII

STRESS ATTACKERS: SER IN THE ATTACK WINDOW (3 SEEDS; THE GATE-AWARE CELLS WITH WEIGHTS 50 AND 500 HAVE 1 SEED). REJ.: REJECTION RATE OF POISONED BLOCKS.

Attacker	Receiver	Pre	Window	Rej.
A2 targeted, Jakes 0.05, SIR 0 dB	Online DeepSIC	0.019	0.438	–
	Meta-DeepSIC	0.017	0.350†	–
	Citadel-Rx, G1+G2	0.010	0.010	0.82
	Citadel-Rx, G1+G2+G3	0.010	0.010	0.83
Gate-aware, SIR 10 dB	Citadel-Rx, no gates	0.039	0.531	0.00
	G1 only, weight 5	0.039	0.034	0.93
	G1 only, weight 50	0.039	0.044	0.85
	G1 only, weight 500	0.039	0.074	0.30
	G1+G2, weight 5	0.039	0.024	1.00
	G1+G2+G3, weight 5	0.039	0.026	0.99
	G1+G2+G3, weight 500	0.039	0.027	0.95
A3 boiling frog, 300 blocks	Citadel-Rx, $C_K = 8$	0.029	0.040‡	0.00
	Citadel-Rx, $C_K = 32$	0.031	0.039‡	0.00
	Citadel-Rx, $C_K = 128$	0.032	0.039‡	0.00

†post-attack 0.053, recovery 23 blocks. ‡post-attack 0.023.

deviation is Δ_f itself shrinking with C_K, because a slower anchor pulls the fast level back harder. The measured fast displacement (16–34) is of the size the geometric sum predicts from L_θ alone ($\epsilon G_\Phi/(1 - L_\theta) \approx 39$), so the anchor coupling does not compound even though $L_\theta + L_a > 1$.

H. Ablations

Each component is removed in turn (Table VIII). Three carry the result. The slow memory: with one level the gates fire but nothing uncorrupted is left to fall back on, and the receiver

TABLE VIII
ABLATIONS UNDER WHITE-BOX A1 AT SIR 0 DB, 3 SEEDS (COMPLETE RECEIVER ON THE SAME SEEDS). SCRIPT: SWEEP.PY ABLATE.

Configuration	Pre	Window	Rej.	False rej.
Complete receiver	0.039	**0.028**	0.97	0.003
$K = 1$ (gates, no slow level)	0.068	0.788	0.98	0.222
$K = 2$, $C = (1, 8)$	0.038	0.050	0.97	0.006
$K = 4$, $C = (1, 4, 16, 64)$	0.038	0.029	0.97	0.011
Periods misaligned, $C = (1, 3, 7)$	0.037	0.031	0.95	0.000
No gates	0.039	0.540	0.00	–
G1 only	0.039	0.031	0.97	0.000
G2 only	0.039	0.028	0.97	0.000
G3 only	0.039	0.507	0.08	0.000
Last layer adaptable	0.052	0.029	0.98	0.011
Plain GD fast rule	0.066	0.048	0.90	0.072
Adam without delta decay	0.040	0.030	0.97	0.006
Fixed step η	0.040	0.027	0.97	0.011
No anchor ($\lambda = 0$)	**0.058**	0.031	0.99	0.178
Jump re-anchor [13], $\beta = 0.25$	0.058	0.031	0.97	0.150
No fallback	0.040	**0.245**	0.97	0.011
Keep rejected update (no revert)	0.040	0.025	1.00	0.017

collapses like online DeepSIC (0.788); two levels halve the damage (0.050), three remove it. The fallback: reverting a rejected update protects the memory but leaves the data of the block to a poisoned fast level (0.245). The proximal anchor: it is the clean gain (0.039 against 0.058 without it), and the jump re-anchoring of [13] is no better than no anchoring; both also raise clean false rejections to 15–18%. The negative results are equally clear: the delta-rule decay and the surprise-driven step each buy 2–3% of clean SER; period alignment does not matter on this channel; G3 is useless against A1, as predicted, its role being A2; and whether a rejected update is reverted or kept makes no difference once the fallback is active.

I. Generalisation: Channels, System Size and Doppler

Table IX repeats the main comparison on the nonlinear channel, the COST2100 traces, an 8×8 system and three Doppler rates. In every case, every baseline is destroyed at SIR ≤ 0 dB and Citadel-Rx stays at or below its pre-attack level. With eight users the clean gain grows (Citadel-Rx 0.102 against LMMSE 0.129 and online DeepSIC 0.204 over 100 clean blocks), Meta-DeepSIC and MedBN again never recover (post-attack 0.40–0.66), and false rejections rise to 3–5% with the deeper eight-user fades. Faster fading raises every receiver's clean SER and the gate's false alarms ($0.6 \to 3.3\%$), not the attacker's success: the attacked SER of Citadel-Rx is flat at 0.007 over a $10\times$ range of Doppler.

J. COST2100 and the Relative Slack

COST2100 at 10 dB is the one setting in which the first version of the receiver left its clean trajectory. With the absolute CRC slack ($\delta = 0.01$), the attack-window SER of Citadel-Rx was $0.0022/0.0032/0.0054/0.0122/0.0111$ at SIR -10 to 10 dB against a clean 0.0003, with rejection falling to 0.64: a one-point slack is $30\times$ the clean SER, so the attacker can pass G2 while doing damage. The relative slack of (10) brings the window to $0.0011/0.0012/0.0012/0.0015/0.0028$ (rejection $0.89/0.90/0.86/0.74/0.40$, no false rejections) and leaves the linear grid unchanged (0.025–0.029, false rejection $\leq 2\%$). A first calibration with a floor of 10^{-3} had fixed COST2100 but raised linear false rejections to 15–20%, because the floor sat inside the statistical resolution of the pass fraction; $2 \cdot 10^{-3}$ is five standard deviations of it. On this nearly diagonal channel LMMSE attacked through the surrogate is itself nearly immune (0.0010 at SIR 0 dB), so Citadel-Rx matches rather than beats the model-based receiver under attack (0.0013; paired difference $+0.0003$, interval $[-0.0005, +0.0010]$).

K. Cost

Per block on one CPU worker (eight single-thread workers in parallel, 300 adaptation steps): LMMSE 0.07 s, online DeepSIC 4.0 s, drift-detection 9.2 s, Meta-DeepSIC 12.4 s, Citadel-Rx 12.2 s, single-memory Titans 14.4 s, MedBN-style 37.2 s. Citadel-Rx costs the same as Meta-DeepSIC; its overhead is the gate (one LMMSE detection and one CRC check per block), while the levels cost memory, not compute. The white-box attacker spends 3.2 s of PGD per poisoned block.

VII. DISCUSSION

Some conclusions are firmer than others. Against the attacks of [7] and their stronger variants, the robustness of the receiver is not in doubt: the attacked SER equals the clean one across budgets, knowledge models, channel families, system sizes and Doppler rates, with intervals and tests that leave no room for a seed artefact. The ablation identifies which parts matter (the slow level, the fallback, and G1 or G2) and which do not (period alignment, the delta decay, the surprise rate, reverting versus keeping a rejected update). This is stated plainly because it is the finding most likely to be misread: the value of the receiver lies in its memory and its fallback, not in a cleverer fast learner.

The clean gain is a second contribution, but a conditional one: large where interference cancellation matters and nil on a nearly diagonal channel where LMMSE is already near optimal. The ablation attributes it to the proximal anchor, a prior learned across blocks, which is why the jump re-anchoring of nested learning does not produce it.

The certificate is the structural bound (9) with a measured fast-level displacement; it holds at every window length and exhibits the $1/C_K$ mechanism. The contraction branch of Theorem 1 is not certified for the default receiver, since the measured $L_\theta + L_a$ exceeds one at every proximal weight; a guarantee with no measured quantity requires a fast learner that is a joint contraction in its state and its anchor, which the present design is not.

Two limits are stated as such. At clean SER near 10^{-4} the attacked window sits near 10^{-3}: the few poisoned blocks that pass the gate set a floor that a different slack does not move. And the CRC gate needs CRC-framed data; with uncoded symbols only G1 and G3 remain, and G1 alone defeats the non-adaptive attacker but not the gate-aware one.

The remaining assumptions bound the generality of the results: a pilot-perturbing attacker with an energy budget (a directional jammer or a data spoofer lies outside the model); the block-fading channel families of the DeepSIC literature, with no link-level simulation under a 3GPP model and LDPC coding; QPSK with 100 pilot symbols per user; and DeepSIC as the only backbone tested, although the levels and gates are agnostic to the per-user network.

TABLE IX
TRANSFER TO OTHER CHANNELS, A LARGER SYSTEM AND FASTER FADING: SER IN THE ATTACK WINDOW UNDER WHITE-BOX A1, 3 SEEDS; PRE-ATTACK SER IN PARENTHESES; CITADEL-RX ON COST2100 WITH THE RELATIVE SLACK. SCRIPTS: `SWEEP.PY E3NL`, `E3REL`, `E8`, `E9`.

	tanh PA, 4×4		COST2100, 4×4		SED, 8×8		Jakes, $f_D T_b = 0.01/0.05/0.10$
Receiver	SIR -10	SIR 0	SIR -10	SIR 0	SIR -10	SIR 0	SIR 0
LMMSE (transfer)	0.841 (0.058)	0.421	0.063 (0.0001)	0.0010	0.983 (0.141)	0.793	0.067 / 0.067 / 0.103 (0.015–0.021)
Online DeepSIC	0.973 (0.079)	0.784	0.928 (0.0008)	0.636	0.998 (0.250)	0.971	0.719 / 0.719 / 0.717 (0.017–0.026)
Meta-DeepSIC	0.865 (0.067)	0.656	0.791 (0.0007)	0.345	0.972 (0.220)	0.934	0.496 / 0.501 / 0.531 (0.013–0.026)
Drift-detection	0.973 (0.083)	0.794	0.941 (0.0005)	0.630	0.997 (0.259)	0.971	0.728 / 0.738 / 0.717 (0.023–0.040)
MedBN-style	0.908 (0.069)	0.706	0.744 (0.0002)	0.505	0.995 (0.158)	0.923	0.711 / 0.692 / 0.691 (0.015–0.025)
Single-memory Titans	0.971 (0.075)	0.799	0.917 (0.0011)	0.627	0.998 (0.234)	0.969	0.738 / 0.749 / 0.736 (0.015–0.023)
Citadel-Rx (proposed)	**0.026** (0.042)	**0.026**	**0.0011** (0.0003)	**0.0013**	**0.080** (0.127)	**0.080**	**0.0066 / 0.0069 / 0.0075** (0.007–0.015)
rejection / false rej.	0.99 / 0.017	0.98 / 0.006	0.89 / 0.00	0.86 / 0.00	1.00 / 0.05	1.00 / 0.05	0.92 / 0.94 / 0.97; 0.006–0.033

VIII. CONCLUSION

This paper presented a DeepSIC receiver that keeps its adaptable parameters at three time-scales, consolidates from fast to slow memory only through blocks that pass three label-free physics-consistency tests, pulls the fast level toward the slow projection, and detects a rejected block with the slow memory. Against the pilot-poisoning attacks that destroy online and meta-learned receivers, the attacked symbol error rate equals the clean one at every budget, on four channel families, two system sizes and three Doppler rates, with 86–99% of poisoned blocks rejected and at most 1.7% of clean ones; without attack the receiver halves the error rate of online DeepSIC where interference cancellation matters. The exposure of the slow memory to a bounded attack is divided by its period and multiplied by the miss rate of the gate, and the bound holds with measured constants. Two lessons follow: the defence lies in the memory, not in the learner, and a gate tied to the attacker's own objective is what a gate-aware attacker cannot evade without giving that objective up. The stated limits point to a re-detection pass for accepted blocks, a jointly contractive fast learner, link-level simulation with LDPC coding, and other backbones, to which the levels and gates apply unchanged.

REFERENCES

[1] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.

[2] M. Honkala, D. Korpi, and J. M. J. Huttunen, "DeepRx: Fully convolutional deep learning receiver," *IEEE Trans. Wireless Commun.*, vol. 20, no. 6, pp. 3925–3940, Jun. 2021.

[3] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, May 2019.

[4] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.

[5] N. Shlezinger, R. Fu, and Y. C. Eldar, "DeepSIC: Deep soft interference cancellation for multiuser MIMO detection," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 1349–1362, Feb. 2021.

[6] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.

[7] K. Wu, W. Jiang, D. Niyato, Y. Li, and C. Luo, "Transfer-based adversarial poisoning attacks for online (MIMO-)deep receivers," arXiv:2409.02430, Sep. 2024.

[8] T. Wu, F. Jia, X. Qi, J. T. Wang, V. Sehwag, S. Mahloujifar, and P. Mittal, "Uncovering adversarial risks of test-time adaptation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023.

[9] H. Park, J. Hwang, S. Mun, S. Park, and J. Ok, "MedBN: Robust test-time adaptation against malicious test samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024.

[10] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.

[11] J. Steinhardt, P. W. Koh, and P. Liang, "Certified defenses for data poisoning attacks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.

[12] T. Lykouris, V. Mirrokni, and R. Paes Leme, "Stochastic bandits robust to adversarial corruptions," in *Proc. ACM Symp. Theory Comput. (STOC)*, 2018.

[13] A. Behrouz, M. Razaviyayn, P. Zhong, and V. Mirrokni, "Nested learning: The illusion of deep learning architectures," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025.

[14] A. Behrouz, P. Zhong, and V. Mirrokni, "Titans: Learning to memorize at test time," arXiv:2501.00663, 2025.

[15] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017.

[16] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[17] M. Zinkevich, "Online convex programming and generalized infinitesimal gradient ascent," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2003.

[18] A. Mokhtari, S. Shahrampour, A. Jadbabaie, and A. Ribeiro, "Online optimization in dynamic environments: Improved regret rates for strongly convex problems," in *Proc. IEEE Conf. Decis. Control (CDC)*, 2016, arXiv:1603.04954.

[19] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *J. Amer. Statist. Assoc.*, vol. 58, no. 301, pp. 13–30, 1963.

[20] W. C. Jakes, *Microwave Mobile Communications*. New York, NY, USA: Wiley, 1974.

[21] L. Liu, C. Oestges, J. Poutanen, K. Haneda, P. Vainikainen, F. Quitin, F. Tufvesson, and P. De Doncker, "The COST 2100 MIMO channel model," *IEEE Wireless Commun.*, vol. 19, no. 6, pp. 92–99, Dec. 2012.

[22] 3GPP, "NR; Multiplexing and channel coding," TS 38.212, 2024.

[23] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[24] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, 1945.

[25] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.